

SPACE PROGRAMME: ISRO, ORGANISATION AND LAUNCH VEHICLES

INSTITUTIONAL COMMAND MAP → CENTRE-TO-LAUNCH CHAIN → POLICY AND LAW STACK → LAUNCHER-PAYLOAD SEPARATION → ORBIT VOCABULARY MATRIX → GTO-TO-GEO RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 INSTITUTIONAL COMMAND MAP

INSTITUTIONAL COMMAND MAP DEPARTMENT OF SPACE POLICY UMBRELLA AND SOVEREIGN DIRECTION PUBLIC R&D, LAUNCH VEHICLES, SPACECRAFT AND MISSIONS PROMOTE, AUTHORISE AND SUPERVISE NON-GOVERNMENT ACTIVITY COMMERCIAL CONTRACTS, SERVICES AND TECHNOLOGY TRANSFER

INSTITUTION NAMES ARE NOT INTERCHANGEABLE

DEPARTMENT OF SPACE → policy umbrella and sovereign direction

ISRO → public R&D, launch vehicles, spacecraft and missions

IN-SPACe → promote, authorise and supervise non-government activity

NSIL → commercial contracts, services and technology transfer

CONCEPT / ARCHITECTURE

- DEPARTMENT OF SPACE → policy umbrella and sovereign direction
- ISRO → public R&D, launch vehicles, spacecraft and missions

MECHANISM / APPLICATION

- IN-SPACe → promote, authorise and supervise non-government activity
- NSIL → commercial contracts, services and technology transfer

READINESS / STATUS LIMIT

- RULE → institution names are not interchangeable
- DEPARTMENT OF SPACE → policy umbrella and sovereign direction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NSIL → commercial contracts, services and technology transfer.

01

STAGE 01 CENTRE-TO-LAUNCH CHAIN

CENTRE-TO-LAUNCH CHAIN LAUNCH-VEHICLE DESIGN AND DEVELOPMENT LIQUID PROPULSION AND PROPULSION TESTING ROLES INTEGRATION, RANGE AND LAUNCH OPERATIONS MISSION CONTROL FLIGHT MONITORING AND DECISION CHAIN A CENTRE FUNCTION IS NOT THE WHOLE ISRO MANDATE

VSSC → launch-vehicle design and development

LPSC / IPRC → liquid propulsion and propulsion testing roles

SDSC-SHAR → integration, range and launch operations

MISSION CONTROL → flight monitoring and decision chain

RULE → a centre function is not the whole ISRO mandate

STARTING CONDITION / INPUT

- VSSC → launch-vehicle design and development
- LPSC / IPRC → liquid propulsion and propulsion testing roles

SCIENTIFIC OR ENGINEERING MECHANISM

- SDSC-SHAR → integration, range and launch operations
- MISSION CONTROL → flight monitoring and decision chain

OUTPUT / FAILURE BOUNDARY

- RULE → a centre function is not the whole ISRO mandate
- VSSC → launch-vehicle design and development

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a centre function is not the whole ISRO mandate.

02

STAGE 02 POLICY AND LAW STACK

POLICY AND LAW STACK OUTER-SPACE TREATIES INTERNATIONAL STATE OBLIGATIONS INDIAN SPACE POLICY 2023 EXECUTIVE ROLE ALLOCATION IN-SPACE NORMS AUTHORISATION AND SUPERVISION INTERFACE COMMERCIAL CONTRACT

OUTER-SPACE TREATIES → international State obligations

INDIAN SPACE POLICY 2023 → executive role allocation

IN-SPACe NORMS → authorisation and supervision interface

COMMERCIAL CONTRACT → service and risk allocation

CONCEPT / ARCHITECTURE

- OUTER-SPACE TREATIES → international State obligations
- INDIAN SPACE POLICY 2023 → executive role allocation

MECHANISM / APPLICATION

- IN-SPACe NORMS → authorisation and supervision interface
- COMMERCIAL CONTRACT → service and risk allocation

READINESS / STATUS LIMIT

- GAP → policy is not a comprehensive domestic space statute
- OUTER-SPACE TREATIES → international State obligations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GAP → policy is not a comprehensive domestic space statute.

03

STAGE 03 LAUNCHER-PAYLOAD SEPARATION

LAUNCHER-PAYLOAD SEPARATION LAUNCH VEHICLE SUPPLIES ASCENT ENERGY AND GUIDANCE PAYLOAD FAIRING PROTECTS THE MISSION PAYLOAD LAUNCHER TASK AND SPACECRAFT TASK DIVERGE CARRIES PAYLOAD AND MAY RAISE OR MAINTAIN ORBIT

ROCKET SUCCESS AND SPACECRAFT SERVICE ENTRY ARE SEPARATE

LAUNCH VEHICLE → supplies ascent energy and guidance

PAYLOAD FAIRING → protects the mission payload

SEPARATION → launcher task and spacecraft task diverge

SPACECRAFT → carries payload and may raise or maintain orbit

RULE → rocket success and spacecraft service entry are separate

03

CONCEPT / ARCHITECTURE

- OUTER-SPACE TREATIES → international State obligations
- INDIAN SPACE POLICY 2023 → executive role allocation

MECHANISM / APPLICATION

- IN-SPACE NORMS → authorisation and supervision interface
- COMMERCIAL CONTRACT → service and risk allocation

READINESS / STATUS LIMIT

- GAP → policy is not a comprehensive domestic space statute
- OUTER-SPACE TREATIES → international State obligations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GAP → policy is not a comprehensive domestic space statute.

04

STAGE 03 LAUNCHER-PAYLOAD SEPARATION

LAUNCHER-PAYLOAD SEPARATION

LAUNCH VEHICLE

SUPPLIES ASCENT ENERGY AND GUIDANCE

PAYLOAD FAIRING

PROTECTS THE MISSION PAYLOAD

LAUNCHER TASK AND SPACECRAFT TASK DIVERGE

CARRIES PAYLOAD AND MAY RAISE OR MAINTAIN ORBIT

ROCKET SUCCESS AND SPACECRAFT SERVICE ENTRY ARE SEPARATE

LAUNCH VEHICLE → supplies ascent energy and guidance

PAYLOAD FAIRING → protects the mission payload

SEPARATION → launcher task and spacecraft task diverge

SPACECRAFT → carries payload and may raise or maintain orbit

RULE → rocket success and spacecraft service entry are separate

STARTING CONDITION / INPUT

LAUNCH VEHICLE → supplies ascent energy and guidance

PAYLOAD FAIRING → protects the mission payload

SCIENTIFIC OR ENGINEERING MECHANISM

SEPARATION → launcher task and spacecraft task diverge

SPACECRAFT → carries payload and may raise or maintain orbit

OUTPUT / FAILURE BOUNDARY

RULE → rocket success and spacecraft service entry are separate

LAUNCH VEHICLE → supplies ascent energy and guidance

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → rocket success and spacecraft service entry are separate.

05

STAGE 04 ORBIT VOCABULARY MATRIX

ORBIT VOCABULARY MATRIX

ALTITUDE CLASS, OFTEN USED BY CREW AND EARTH OBSERVATION

HIGH-INCLINATION GROUND-TRACK GEOMETRY

REPEATED LOCAL-SOLAR-TIME CROSSING

ORBITAL PERIOD TIED TO EARTH'S ROTATION

CIRCULAR EQUATORIAL GEOSYNCHRONOUS SPECIAL CASE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
LEO → altitude class, often used by crew and Earth observation	SUN-SYNCHRONOUS → repeated local-solar-time crossing	GEO → circular equatorial geosynchronous special case
POLAR → high-inclination ground-track geometry	GSO → orbital period tied to Earth's rotation	LEO → altitude class, often used by crew and Earth observation

SYSTEM / DEVICE / INSTITUTION

LEO → altitude class, often used by crew and Earth observation

POLAR → high-inclination ground-track geometry

DECISIVE TECHNICAL DISTINCTION

SUN-SYNCHRONOUS → repeated local-solar-time crossing

GSO → orbital period tied to Earth's rotation

STATUS / UNIT / EVIDENCE LIMIT

GEO → circular equatorial geosynchronous special case

LEO → altitude class, often used by crew and Earth observation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GSO → orbital period tied to Earth's rotation.

06

STAGE 05 GTO-TO-GEO RAIL

GTO-TO-GEO RAIL

LAUNCHER ASCENT

REACHES TRANSFER CONDITIONS

GTO INJECTION

ELLIPTICAL INTERMEDIATE ORBIT

SPACECRAFT PROPULSION

RAISES PERIGEE AND CIRCULARISES

GEO SLOT

LAUNCHER ASCENT → reaches transfer conditions

GTO INJECTION → elliptical intermediate orbit

SPACECRAFT PROPULSION → raises perigee and circularises

GEO SLOT → circular equatorial final service orbit

TRAP → GTO delivery is not direct GEO placement

STARTING CONDITION / INPUT

LAUNCHER ASCENT → reaches transfer conditions

GTO INJECTION → elliptical intermediate orbit

SCIENTIFIC OR ENGINEERING MECHANISM

SPACECRAFT PROPULSION → raises perigee and circularises

GEO SLOT → circular equatorial final service orbit

OUTPUT / FAILURE BOUNDARY

TRAP → GTO delivery is not direct GEO placement

LAUNCHER ASCENT → reaches transfer conditions

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → GTO delivery is not direct GEO placement.

07

STAGE 06 OPERATIONAL LAUNCHER MATRIX

OPERATIONAL LAUNCHER MATRIX

VERSATILE WORKHORSE AND MIXED MISSION PROFILES

CRYOGENIC UPPER STAGE AND TRANSFER-ORBIT ROLE

HEAVIER OPERATIONAL CLASS AND HUMAN-RATING BASE

SEPARATE SMALL-SATELLITE LAUNCH ARCHITECTURE

ROLE, PROPULSION, ORBIT AND STATUS MUST TRAVEL TOGETHER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PSLV → versatile workhorse and mixed mission profiles	LVM3 → heavier operational class and human-rating base	RULE → role, propulsion, orbit and status must travel together
GSLV → cryogenic upper stage and transfer-orbit role	SSLV → separate small-satellite launch architecture	PSLV → versatile workhorse and mixed mission profiles

SYSTEM / DEVICE / INSTITUTION

PSLV → versatile workhorse and mixed mission profiles

DECISIVE TECHNICAL DISTINCTION

LVM3 → heavier operational class and human-rating base

STATUS / UNIT / EVIDENCE LIMIT

RULE → role, propulsion, orbit and status must travel together

SPACE PROGRAMME: ISRO, ORGANISATION AND LAUNCH VEHICLES • TILE 2/4 • same master rows 2504-5738 of 10746 • continues below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → GTO delivery is not direct GEO placement.

06

STAGE 06 OPERATIONAL LAUNCHER MATRIX

OPERATIONAL LAUNCHER MATRIX

VERSATILE WORKHORSE AND MIXED MISSION PROFILES

CRYOGENIC UPPER STAGE AND TRANSFER-ORBIT ROLE

HEAVIER OPERATIONAL CLASS AND HUMAN-RATING BASE

SEPARATE SMALL-SATELLITE LAUNCH ARCHITECTURE

ROLE, PROPULSION, ORBIT AND STATUS MUST TRAVEL TOGETHER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PSLV → versatile workhorse and mixed mission profiles	LVM3 → heavier operational class and human-rating base	RULE → role, propulsion, orbit and status must travel together
GSLV → cryogenic upper stage and transfer-orbit role	SSLV → separate small-satellite launch architecture	PSLV → versatile workhorse and mixed mission profiles
SYSTEM / DEVICE / INSTITUTION - PSLV → versatile workhorse and mixed mission profiles - GSLV → cryogenic upper stage and transfer-orbit role	DECISIVE TECHNICAL DISTINCTION - LVM3 → heavier operational class and human-rating base - SSLV → separate small-satellite launch architecture	STATUS / UNIT / EVIDENCE LIMIT - RULE → role, propulsion, orbit and status must travel together - PSLV → versatile workhorse and mixed mission profiles

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → role, propulsion, orbit and status must travel together.

07

STAGE 07 LINEAGE AND STATUS LADDER

LINEAGE AND STATUS LADDER

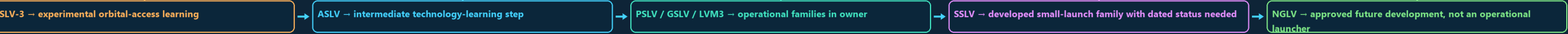
EXPERIMENTAL ORBITAL-ACCESS LEARNING

INTERMEDIATE TECHNOLOGY-LEARNING STEP

OPERATIONAL FAMILIES IN OWNER

DEVELOPED SMALL-LAUNCH FAMILY WITH DATED STATUS NEEDED

APPROVED FUTURE DEVELOPMENT, NOT AN OPERATIONAL LAUNCHER



STARTING CONDITION / INPUT

- SLV-3 → experimental orbital-access learning
- ASLV → intermediate technology-learning step

SCIENTIFIC OR ENGINEERING MECHANISM

- PSLV / GSLV / LVM3 → operational families in owner
- SSLV → developed small-launch family with dated status needed

OUTPUT / FAILURE BOUNDARY

- NGLV → approved future development, not an operational launcher
- SLV-3 → experimental orbital-access learning

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NGLV → approved future development, not an operational launcher.

08

STAGE 08 PROPULSION COMPARISON

PROPULSION COMPARISON

STORED PROPELLANT AND LIMITED THROTTLE CONTROL

EARTH-STORABLE LIQUID

AMBIENT-TEMPERATURE LIQUID PAIR

LIQUID OXYGEN PLUS LIQUID HYDROGEN

LIQUID OXYGEN PLUS KEROSENE

PROPELLANT CLASS DOES NOT PROVE FLIGHT STATUS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SOLID → stored propellant and limited throttle control	CRYOGENIC → liquid oxygen plus liquid hydrogen	RULE → propellant class does not prove flight status
EARTH-STORABLE LIQUID → ambient-temperature liquid pair	SEMI-CRYOGENIC → liquid oxygen plus kerosene	SOLID → stored propellant and limited throttle control
SYSTEM / DEVICE / INSTITUTION - SOLID → stored propellant and limited throttle control - EARTH-STORABLE LIQUID → ambient-temperature liquid pair	DECISIVE TECHNICAL DISTINCTION - CRYOGENIC → liquid oxygen plus liquid hydrogen - SEMI-CRYOGENIC → liquid oxygen plus kerosene	STATUS / UNIT / EVIDENCE LIMIT - RULE → propellant class does not prove flight status - SOLID → stored propellant and limited throttle control

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → propellant class does not prove flight status.

09

STAGE 09 HUMAN-RATING GATE

HUMAN-RATING GATE

RELIABLE LAUNCHER

NECESSARY BUT INSUFFICIENT BASE

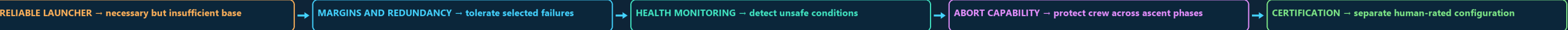
MARGINS AND REDUNDANCY

TOLERATE SELECTED FAILURES

HEALTH MONITORING

DETECT UNSAFE CONDITIONS

ABORT CAPABILITY



STARTING CONDITION / INPUT

- RELIABLE LAUNCHER → necessary but insufficient base
- MARGINS AND REDUNDANCY → tolerate selected failures

SCIENTIFIC OR ENGINEERING MECHANISM

- HEALTH MONITORING → detect unsafe conditions
- ABORT CAPABILITY → protect crew across ascent phases

OUTPUT / FAILURE BOUNDARY

- CERTIFICATION → separate human-rated configuration
- MUST REMEMBER: India's space architecture separates the Department of Space, ISRO...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: India's space architecture separates the Department of Space, ISRO.

